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Robust Deep-Space Navigation Via Four-Body Angles-Only Analysis

SPACE
LABORATORY

Introduction

Angles-only navigation is an optical trajectory estimation method that uses line-of-sight measurements of known celestial bodies in the solar system. It can reduce reliance on ground-based tracking and does not require direct range measurements, making it suitable for autonomous deep-space navigation. Because the measurements can be obtained using passive optical sensors, the method also offers a relatively simple measurement and calculation architecture for onboard implementation.

However, a single three-body analysis using the Sun as a reference and two other planets may result in a sign ambiguity, as the reconstructed spacecraft position may lie either above or below the plane defined by the Sun and the two planets. Resolving this ambiguity typically requires additional orbital information or prior knowledge of the spacecraft trajectory. To avoid this dependency on additional information, this research extends the method to a redundant four-body formulation that includes the Sun and three observed planets.

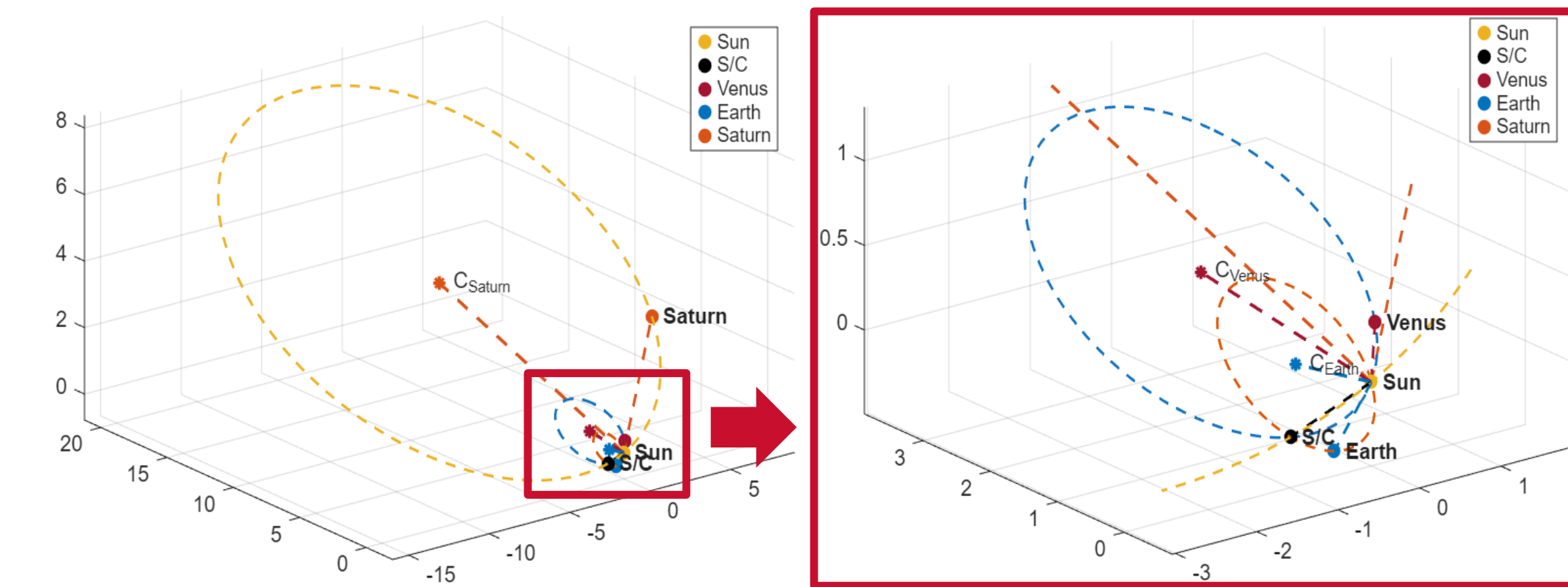


Fig. 1: Representative four-body geometry for analysis (04/22/2029)

Methods

The four-body method introduces redundancy into angles-only navigation. Rather than relying on external orbital information to resolve the sign ambiguity inherent in a three-body solution, the method uses consistency among multiple independent three-body subsets. The resulting position candidates are evaluated through minimum-cost consistency selection, medoid-based outlier rejection, and geometry-weighted fusion.

Ambiguity Resolution Methodology

- Input:** Noisy line-of-sight measurements of the Sun and three planets are generated using known ephemerides
- Decomposition:** Three independent Sun-centered three-body subsets are formed, each producing two mirrored position candidates
- Consistency selection:** Solution Candidates are compared & most mutually consistent set is selected using a cost function
- Outlier rejection:** Estimates that deviate strongly from the medoid solution are removed
- Weighted Fusion:** The remaining estimates are combined using geometry-based weights from α , B , and θ . Where α is the Sun-S/C-planet angle, B is the angle between planes, and θ is the Sun-centered angular separation between planets

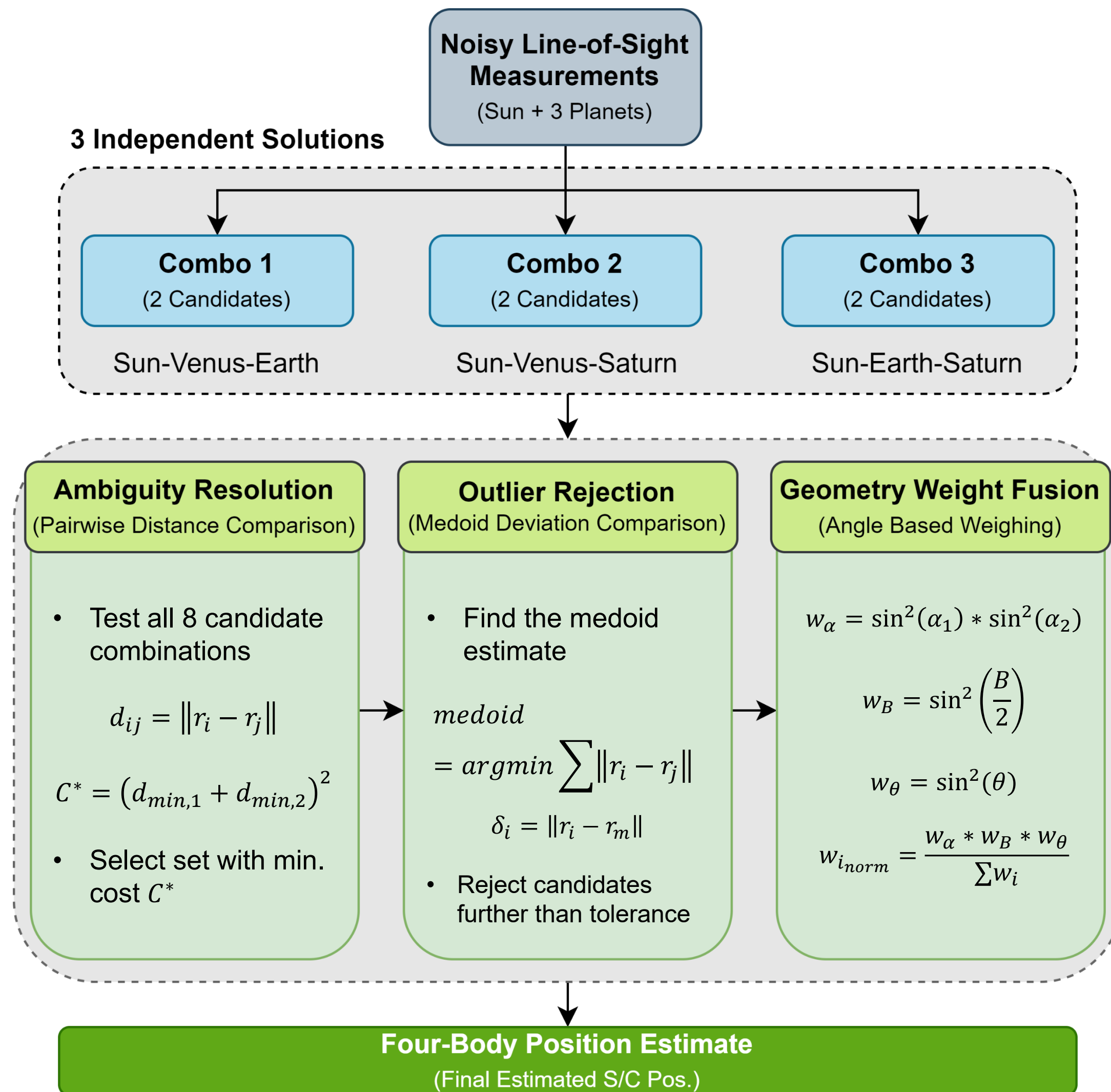


Fig. 2: Four-body angles-only analysis algorithm diagram

Simulation Setup

Sun, Venus, Earth, and Saturn Combinations are used, with the Sun being the reference center. Planetary positions are obtained from the JPL DE405 ephemerides via MATLAB Toolbox. At the representative epoch of April 22nd, 2029, 50,000 Monte Carlo simulations were performed using a camera model corresponding to a $10.84^\circ \times 8.68^\circ$ field-of-view (FOV), and a 1280×1024 pixel resolution. Uniform angular errors of approximately $\pm 0.004234^\circ$ (15.24 arcsec) were applied to the line-of-sight measurements, while the Sun measurements were expected to be provided by a dedicated Sun sensor.

A separate 120-day scan beginning April 1st, 2029, was conducted using 5,000 Monte Carlo trials per day to assess sensitivity to changing observation geometry.

Results

At the representative geometry on April 22nd, 2029, the four-body method achieved an RMSE of 18,771 km, compared to 19,450 km for the best-performing three-body subset, the Sun-Earth-Saturn combination. This corresponds to a 3.5% reduction in RMSE, with additional reductions of 2.4% at the 95th percentile and 5.2% at the 99th percentile. This solution also resolved the sign ambiguity successfully, resulting in a single candidate S/C position estimate. Compared with the weakest subset, the Sun-Venus-Saturn combination, the four-body method reduced the RMSE by approximately 85.6%. The error distribution in Figure 3 shows that the four-body solution remained concentrated near the best-performing Sun-Earth-Saturn subset while substantially reducing the large spread associated with weaker geometries. Figure 4 shows lower exceedance probabilities in the upper error tail relative to the weaker subsets, showing improved robustness.

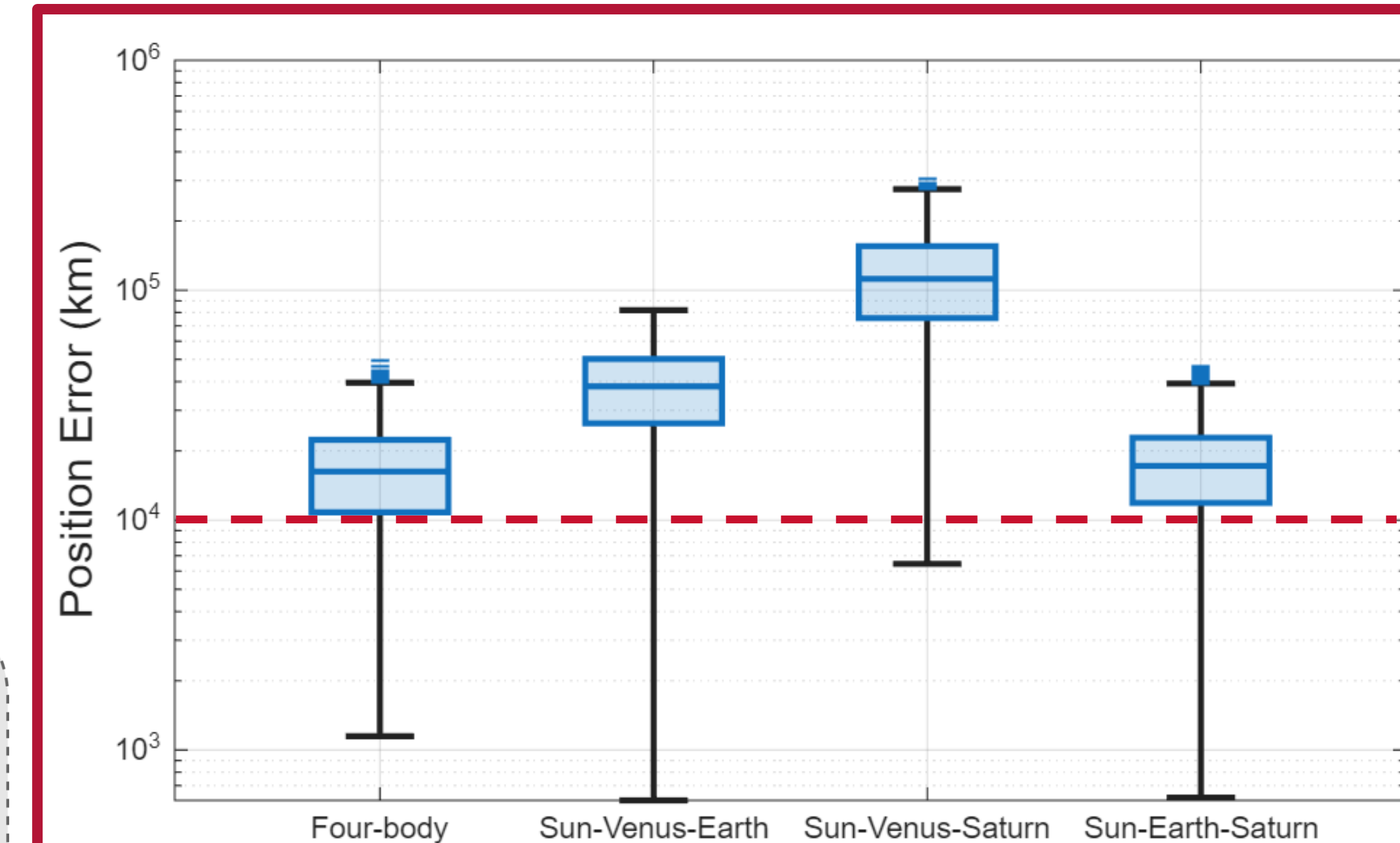


Fig. 3: Position error distribution box chart (04/22/2029)

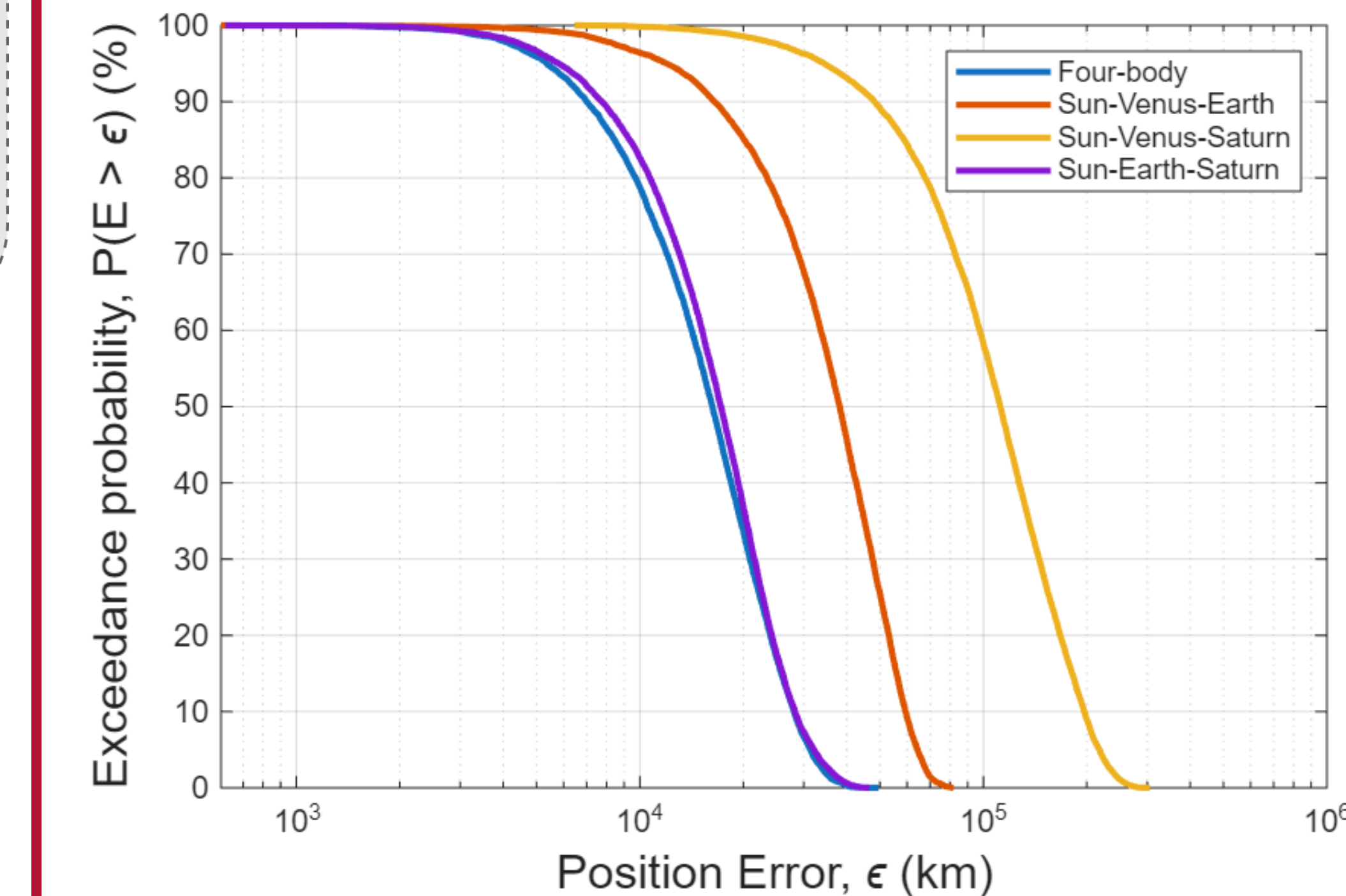


Fig. 4: Position Error Exceedance Probability (04/22/2029)

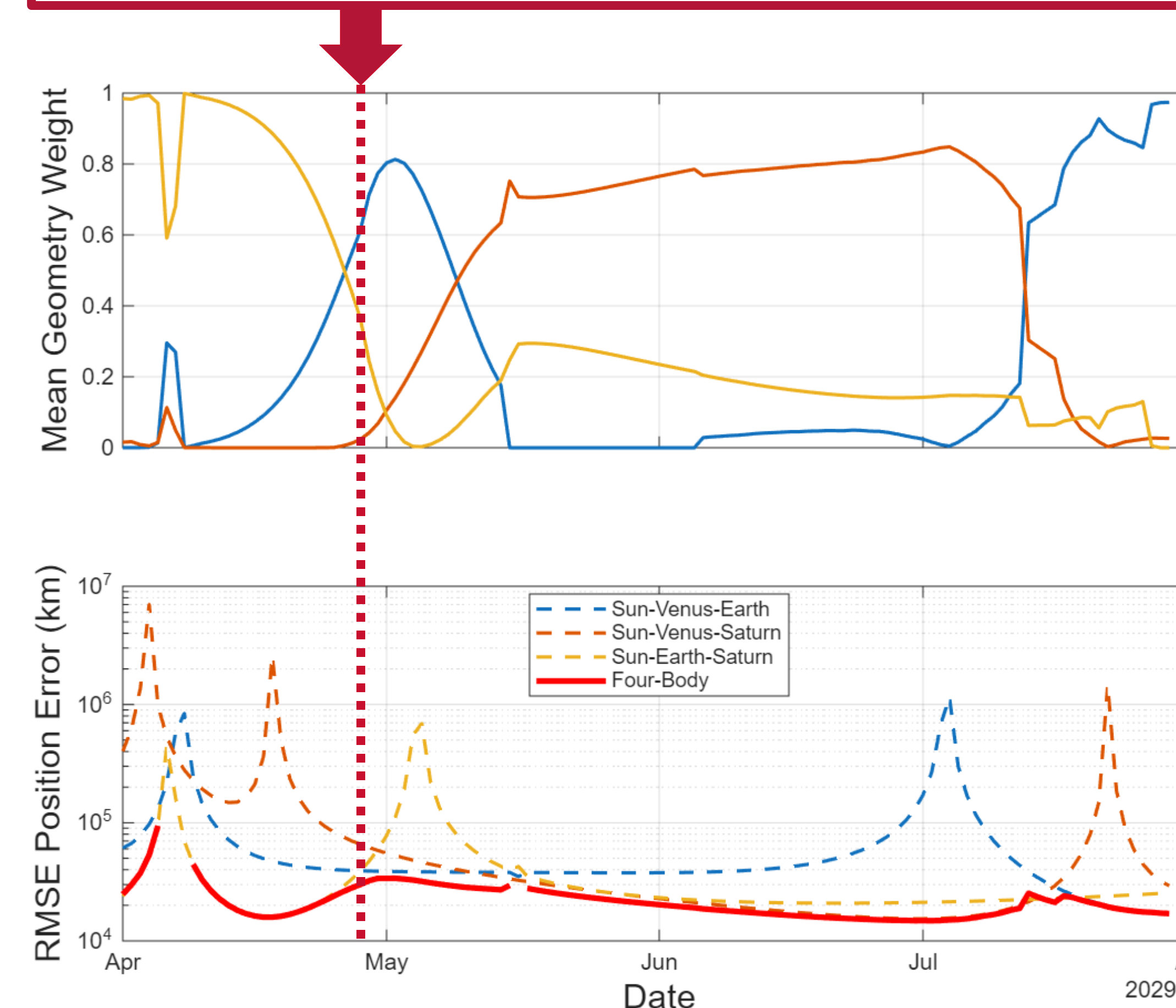


Fig. 5: Geometry weight and RMSE time history over 120-day scan

Case	RMSE (km)	P95 (km)	P99 (km)
Four-Body	18,771	31,348	36,665
Sun-Venus-Earth	41,376	64,237	71,925
Sun-Venus-Saturn	130,084	217,787	251,039
Sun-Earth-Saturn	19,450	32,115	38,579

Table 1: RMSE, P95, and P99 Comparison table (04/22/2029)

Throughout the 120-day scan, the four-body solution generally maintained a lower, more stable RMSE than the individual three-body subsets and was close to the best-performing subsets. Complete or conditional success was achieved in 82.7% of cases, while complete failure and conditional failure occurred in only 17.3% of cases. Failures primarily occurred when multiple subsets simultaneously produced incorrect solution branches.

Outcome	Description	Mean Occurrence (%)
Full Success	All three combinations produce good solutions	45.8
Cond. Success	Two out of three combinations produce good solution	36.9
Cond. Failure	Only one of the three combinations produces a good solution	12.8
Full Failure	All three combinations produce incorrect solutions	4.54

Table 2: Mean solution-outcome classification over the 120-day scan

Conclusion

This research presents a four-body, angles-only navigation algorithm using the Sun and three planets. The method shows that the sign ambiguity in angles-only deep-space navigation could be resolved without external orbital information. The four-body approach improves robustness by rejecting inconsistent solutions and reducing the influence of weak observation geometries, which can also improve estimation accuracy. At the representative geometry, the method remained competitive with the best-performing three-body subset while substantially reducing the influence of poorly conditioned subsets.

However, performance remains geometry-dependent, particularly when multiple subsets select incorrect mirror solution sets. In such cases, an alternative set of observable celestial bodies may provide a more favorable, feasible geometry and restore reliable ambiguity resolution. This suggests that practical implementation could include geometry screening and adaptive selection of the best available body combination.

Future work will develop a fully coupled formulation that solves directly from all four observed bodies and will expand the available observation set using Jupiter Trojan asteroids through JPL SPICE ephemerides.

References

- [1] Choi, S., Negrete, A., Abdelkhalik, O., Servadio, S., and Lee, D. D., "Ambiguity-Free Geometric Approach to Deep-Space Navigation Using Angles-Only Measurements", JGCD, 2026
- [2] JPL DE405 ephemerides via MATLAB Planet Ephemeris